Customer Offer Engagement Dashboard

Part A – One-page User Guide · Prepared by Mahesh Kumar Gupta · October 2026

What it is: a five-page Power BI report plus a Validation page (*02_Final_Customer_Offer_Engagement_Dashboard_Capstone.pbix*) showing how customers respond to promotional offers (received → viewed → completed) and how offer completion relates to spending. It covers 17,000 customers, 10 offers and 306,534 events over a 30-day observation window.

1. Open the report

- **Power BI Desktop:** double-click *02_Final_Customer_Offer_Engagement_Dashboard_Capstone.pbix*. All pages work without the data file; the data file is only needed to refresh.
- **Power BI Service:** open the shared workspace link (or app) and select the report.

2. Refresh the data

- **Desktop:** Home → **Refresh**. If the data file has moved: Home → Transform data → **Data source settings** → Change Source → Browse → Close → Refresh.
- **Service:** refresh runs on the owner's schedule; use **Refresh now** on the semantic model for an immediate update.
- After every refresh, check the Executive Overview cards: Total Customers **17,000** · Offers Received **76,277** · Redemption Rate **55.01%**.

3. Navigate the pages

Page	Question it answers	Main visuals
1 Executive Overview	How are offers performing overall?	Campaign engagement and business performance KPI cards
2 Offer Funnel Chart	Where do offers drop off?	Rate cards, funnel Received → Viewed → Completed, key insight
3 Customer Segmentation	Which customers redeem most?	4 slicers, redemption by age group and by gender, age × income heatmap, key insight
4 Offer Performance Dashboard	Which offer types, rewards and channels work?	Bars by type / reward / channel, difficulty scatter, key insight
5 Transaction Behavior	Do completers spend more?	Avg transaction by day, 30-day spend, spend before vs after
6 Validation	Are the numbers correct?	DAX vs Power Query counts for 100 customers; mismatch card

4. Use slicers and filters

- On **Customer Segmentation**, open a slicer (Gender, Age Group, Income Group, Engagement Segment) and pick values – the charts update.
- Hold **Ctrl** to pick several values; use the **eraser** icon to clear a slicer. Click a bar or cell to cross-filter; click again to clear.
- Hover for exact values; use **Focus mode** to enlarge a visual.
- The key-insight boxes describe the **unfiltered** population and do not change with slicers.

5. Reading the key metrics

- **Redemption Rate** = offers completed ÷ BOGO and discount offers received (informational offers cannot be completed, so they are excluded).
- **View Rate** = viewed ÷ received. **View to Completion Rate** = completed ÷ viewed. **Unknown** = 2,175 customers with no profile data.

6. Key findings at a glance

- **24% of offers are never viewed**; offers sent through social reach a ~93% view rate vs ~73% on web.
- **Redemption 55.01%**: discount 58.6% vs BOGO 51.4%; easy, longer offers work best (DISCOUNT 10/2 10d: 70.0%).
- **Income drives redemption**: 44.5% (30–50k) → 76.8% (90k+); customers with missing profiles redeem only 14.5%.
- **Completers spend ~4× more per transaction** ($14.38 vs $3.67) – an association, not cause.

Part B – Technical Document

Project files, data quality, data model, Power Query, DAX, report inventory, validation and maintenance

B1. Project files

File	Purpose
customers.csv, events.csv, offers.csv, data_dictionary.csv	Raw source data supplied with the brief
01_Data_set_cleaned.xlsx	Cleaned, model-ready dataset loaded by the PBIX – 5 sheets: Customers (17,000, with derived bands, segments and spend fields), Events (306,534, payload parsed), Offers (10), OfferChannels (33), DataDictionary
02_Final_Customer_Offer_Engagement_Dashboard_Capstone.pbix	Power BI report and semantic model: 8 tables, 19 measures (17 report + 2 QA), 5 report pages + Validation page
03_Final_Customer_Offer_Engagement_Dashboard_Capstone.pdf	PDF export of the report: five analysis pages + Validation page
04_Documentation_and_Deployment_v5.pdf	This document – user guide, technical document, deployment guide
05 / 06 Customer_Offer_Engagement_Presentation (.pdf / .pptx)	Presentation deck: findings, testing, recommendations, dashboard appendix
07_Strategy (.pdf / .pptx)	Five strategic recommendations and the decisions needed from leadership

B2. Solution architecture

Stage	Description
1. Source	Raw CSV files (customers, events, offers, data dictionary)
2. Cleaning	Event payloads parsed, demographics standardised, offer channels split, 12 data-quality checks run
3. Power Query	Load tables, promote headers, set data types; FunnelStages and _Measures tables added
4. Model	Star schema: Events fact; Customers and Offers dimensions; OfferChannels bridge
5. Report	5 report pages + Validation page, 19 DAX measures, 4 slicers, conditional-format heatmap, insight boxes
6. Service	Publish, role-based access, scheduled refresh

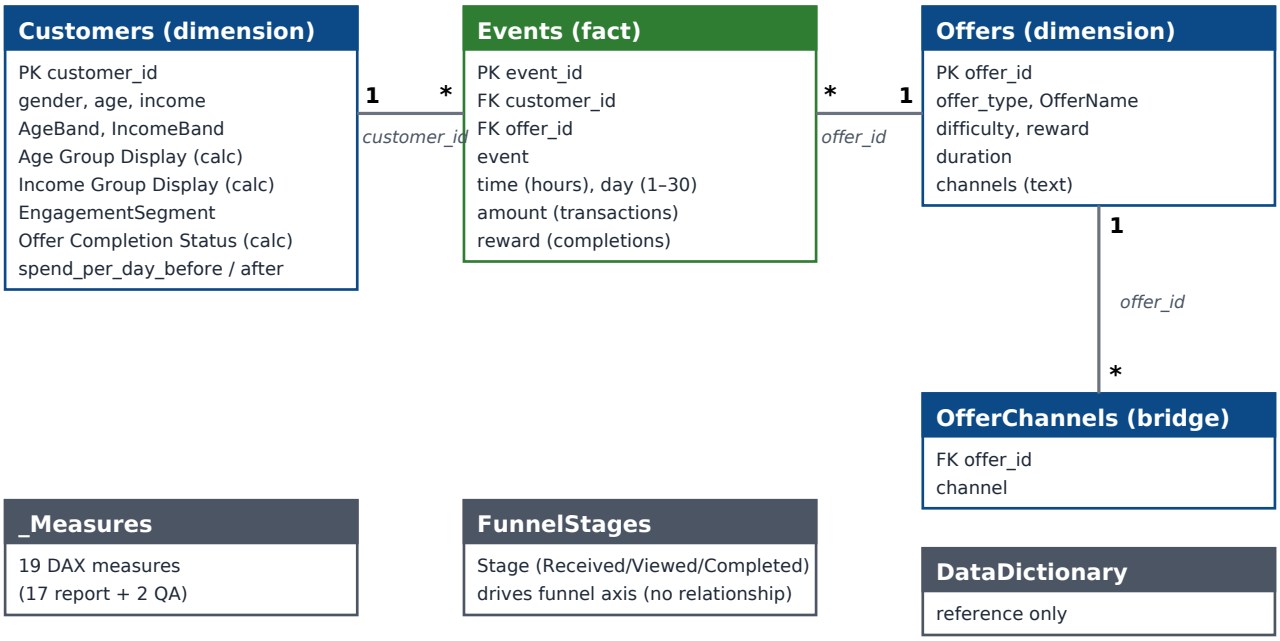
B3. Data quality checks (run on the raw files and 01_Data_set_cleaned.xlsx)

Table	Rows	Check	Issues	Treatment
Customers	17,000	Duplicate customer IDs	0	–
Customers	17,000	Unknown ages	2,175	Placeholder age 118 set to blank; AgeBand null → "Unknown" in Power Query (5 customers aged 101 kept in 65+)
Customers	17,000	Undisclosed genders	2,175	Labelled "Unknown"; M / F / O mapped to Male / Female / Other
Customers	17,000	Missing income (not imputed)	2,175	Kept blank; IncomeBand null → "Unknown" in Power Query
Customers	17,000	Invalid membership dates	0	became_member_on converted to member_date (Date) and member_year
Offers	10	Duplicate offer IDs	0	channels list split into OfferChannels (33 rows)
Events	306,534	Unparseable event payloads	0	value parsed; 'offer id' and 'offer_id' keys unified
Events	306,534	Unknown customer references	0	Referential integrity confirmed
Events	306,534	Unknown offer references	0	Referential integrity confirmed
Events	306,534	Time outside 0–719 hours	0	time kept as authoritative hours; day = time ÷ 24 + 1
Events	306,534	Missing offer ID on offer events	0	–

Table	Rows	Check	Issues	Treatment
Events	306,534	Missing amount on transactions	0	amount numeric for transactions only

Other decisions: informational offers are excluded from redemption denominators; 397 duplicate "offer completed" rows are kept as valid (one purchase can complete two copies of an offer); event time is analysed as day 1–30 of the observation window, not as a calendar date.

B4. Data model (ERD)



From (1)	To (*)	Key	Notes
Customers	Events	customer_id	Every event belongs to one customer
Offers	Events	offer_id	Transactions have a blank offer_id
Offers	OfferChannels	offer_id	Bridge for channel analysis (channel measures apply the channel filter to offers)
Customers	QA_EventCounts	customer_id	One-to-one; used only by the Validation page
FunnelStages, _Measures, DataDictionary	–	–	Disconnected helper tables

B5. Power Query M – JSON extraction for events

Rebuilds the Events table directly from the raw CSV (Home → New Source → Blank Query → Advanced Editor). Replace the file path. Text functions are used because the payload uses single quotes and is not valid JSON.

```
let
    Source   = Csv.Document(File.Contents("C:\PowerBI\events.csv"),
                  [Delimiter=",", Encoding=65001, QuoteStyle=QuoteStyle.Csv]),
    Headers  = Table.PromoteHeaders(Source, [PromoteAllScalars=true]),
    AddOffer = Table.AddColumn(Headers, "offer_id", each
                  if Text.Contains([value], "offer")
                  then Text.BetweenDelimiters([value], ": '", "'") else null, type text),
    AddAmt   = Table.AddColumn(AddOffer, "amount", each
                  if Text.Contains([value], "amount")
                  then Number.From(Text.BetweenDelimiters([value], "'amount': ", "}"), "en-US")
                  else null, type number),
    AddRew   = Table.AddColumn(AddAmt, "reward", each
                  if Text.Contains([value], "reward")
                  then Number.From(Text.BetweenDelimiters([value], "'reward': ", "}"), "en-US")
                  else null, Int64.Type),
    Typed    = Table.TransformColumnTypes(AddRew, {{"time", Int64.Type}}),
    AddDay   = Table.AddColumn(Typed, "day", each Number.IntegerDivide([time], 24) + 1, Int64.Type),
    AddIdx   = Table.AddIndexColumn(AddDay, "event_id", 1, 1, Int64.Type),
    Result   = Table.RemoveColumns(AddIdx, {"value"})
```

```
in
    Result
```

Query	Applied steps
All tables	Source → Navigation (sheet) → Use First Row as Headers → Changed type
Events	amount = Decimal Number; reward, time, day, event_id = Whole Number
Customers	age = Whole Number; income and spend columns = Decimal; dates = Date; AgeBand / IncomeBand null → "Unknown"
Offers	reward and difficulty: Summarization = Don't summarize
OfferChannels, DataDictionary	Use First Row as Headers

B6. Calculated columns and helper tables

Object	Purpose
`Customers[Age Group Display]`	Age group label used on the age chart and heatmap rows (ordered 18-24 … 65+, Unknown)
`Customers[Income Group Display]`	Income group label used on heatmap columns (30-50k … 90k+, Unknown)
`Customers[Offer Completion Status]`	"Completer" if the customer completed at least one offer, otherwise "Non-completer"
`Customers[EngagementSegment]`	High ≥ 60% · Medium 30–60% · Low < 30% completion; No response (none viewed); No offers
`FunnelStages (table)`	Stage values Received / Viewed / Completed with a sort order – the funnel axis
`QA_EventCounts (table)`	Power Query Group By of Events per customer and event type (PQ_Received, PQ_Viewed, PQ_Completed, PQ_Transactions) – independent check
`_Measures (table)`	Container for all 19 measures

B7. DAX measures (table: _Measures)

Measure names are taken from the PBIX report. Formulas show the logic implemented; confirm exact syntax in Power BI (Data pane → click a measure → formula bar).

Measure	Logic / DAX	Used on
Total Customers	`COUNTROWS(Customers)`	P1
Offers Received	`CALCULATE(COUNTROWS(Events), Events[event] = "offer received")`	P1, P4 tip
Offers Viewed	`CALCULATE(COUNTROWS(Events), Events[event] = "offer viewed")`	P1
Offers Completed	`CALCULATE(COUNTROWS(Events), Events[event] = "offer completed")`	P1, P4 tip
Total Transactions	`CALCULATE(COUNTROWS(Events), Events[event] = "transaction")`	P2
Redeemable Offers Received	`CALCULATE([Offers Received], KEEPFILTERS(Offers[offer_type] <> "informational"))`	P4 tip
Redemption Rate	`DIVIDE([Offers Completed], [Redeemable Offers Received])` — labelled "Overall Redemption Rate" on charts	P1–P4
View Rate	`DIVIDE([Offers Viewed], [Offers Received])`	P2
View to Completion Rate	`DIVIDE([Offers Completed], [Offers Viewed])`	P2
Funnel Event Count	`SWITCH(SELECTEDVALUE(FunnelStages[Stage]), "Received", [Offers Received], "Viewed", [Offers Viewed], "Completed", [Offers Completed])`	P2 funnel
Channel Redeemable Offers Received	`Redeemable offers received, filtered to offers sent on the selected channel (via OfferChannels)`	P4
Channel Offers Completed	`Offers completed, filtered to offers sent on the selected channel`	P4
Channel Redemption Rate	`DIVIDE([Channel Offers Completed], [Channel Redeemable Offers Received])`	P4
Average Transaction Amount	`DIVIDE(SUM(Events[amount]), [Total Transactions])`	P1, P5
30-Day Spend per Customer	`DIVIDE(SUM(Events[amount]), [Total Customers])`	P5
Avg Spend per Day Before	`AVERAGE(Customers[spend_per_day_before])`	P5

Measure	Logic / DAX	Used on
Avg Spend per Day After	`AVERAGE(Customers[spend_per_day_after])`	P5
QA Check	`IF(each DAX count = SUM of its QA_EventCounts column, "✔ Match", "✘ Mismatch")`	P6
QA Mismatches	`COUNTROWS(FILTER(VALUES(Customers[customer_id]), [QA Check] = "✘ Mismatch")) + 0`	P6

Design note – KEEPFILTERS: without it, the offer_type filter inside Redeemable Offers Received overrides the offer_type on a chart axis, so each bar is divided by all 61,042 redeemable offers (BOGO would show 25.7% instead of 51.4%).

B8. Report inventory (from the PBIX)

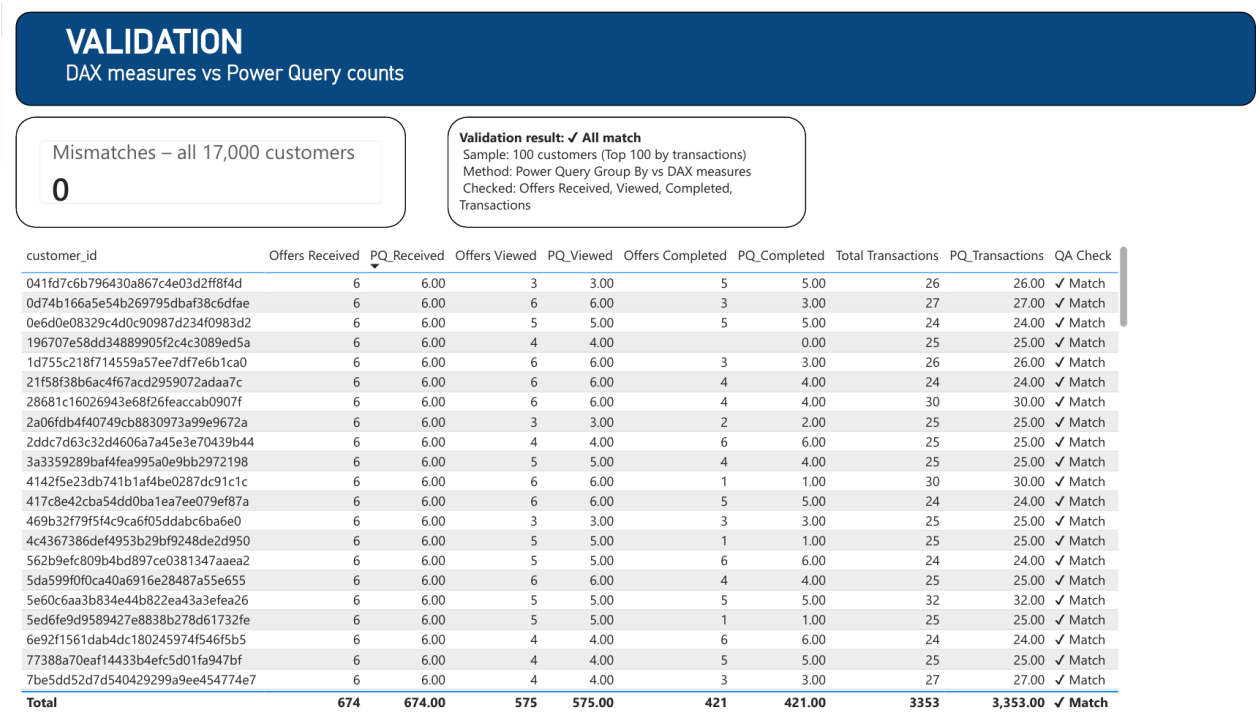
Page	Visuals and fields
1 Executive Overview	6 KPI cards: Total Customers, Offers Received, Offers Viewed, Offers Completed, Redemption Rate, Average Transaction Amount; section headers
2 Offer Funnel Chart	4 cards: Total Transactions, View Rate, Redemption Rate, View to Completion Rate · Funnel: FunnelStages[Stage] × Funnel Event Count · Key insight
3 Customer Segmentation	Slicers: gender, AgeBand, IncomeBand, EngagementSegment · Column chart: Redemption Rate by Age Group Display · Bar chart: Redemption Rate by gender · Matrix heatmap: Age Group Display × Income Group Display · Key insight
4 Offer Performance Dashboard	Bars: Redemption Rate by offer_type and by reward · Bar: Channel Redemption Rate by channel · Scatter: average difficulty × Redemption Rate, size = average reward, by OfferName / offer_type · Key insight
5 Transaction Behavior	Line: Average Transaction Amount by day × Offer Completion Status · Column: 30-Day Spend per Customer by status · Column: Avg Spend per Day Before vs After · Key insight
6 Validation	Card: QA Mismatches across all 17,000 customers · Method text box · Table: top 100 customers by transactions, DAX vs Power Query counts with QA Check

B9. Validation (dashboard vs independent calculation from raw files)

Check	Expected (raw data)	Dashboard	Status
Total customers	17,000	17,000	Pass
Offers received / viewed / completed	76,277 / 57,725 / 33,579	Same	Pass
Total transactions	138,953	139K	Pass
Overall redemption rate	33,579 ÷ 61,042 = 55.01%	55.01%	Pass
View rate / view to completion rate	75.68% / 58.17%	Same	Pass
Redemption by offer type	Discount 58.64% · BOGO 51.38%	Same	Pass
Redemption by reward	$3 67.43% · $2 61.29% · $5 52.67% · $10 46.02%	Same	Pass
Channel redemption	Social 57.24% · Mobile 56.51% · Web 55.99% · Email 55.01%	Same	Pass
Redemption by gender	Female 70.61% · Other 69.49% · Male 53.88% · Unknown 14.48%	Same	Pass
Heatmap 55-64 × 90k+	77.44%	77.44%	Pass
Average transaction amount	$12.78	$12.78	Pass
30-day spend: completers / non-completers	$133 / $18	Same	Pass
Spend per day before / after first completion	$1.17 / $6.27	Same	Pass
100-customer test (Validation page)	Totals 674 / 575 / 421 / 3,353 (received / viewed / completed / transactions)	Same; ✔ Match on all 100 rows	Pass
Full-population check	0 mismatches across 17,000 customers	0	Pass

Expected values were calculated independently from the raw CSV files in Python and compared with the final dashboard pages. The Validation page repeats the check inside Power BI: Power Query counts vs DAX measures for 100 customers, plus a mismatch count across all 17,000.

Evidence – Validation page (from the final report)



customer_id	Offers Received	PQ_Received	Offers Viewed	PQ_Viewed	Offers Completed	PQ_Completed	Total Transactions	PQ_Transactions	QA Check
041fd7c6b796430a867c4e03d2ff8f4d	6	6.00	3	3.00	5	5.00	26	26.00	✓ Match
0d74b166a5e54b269795dbaf38c6dfae	6	6.00	6	6.00	3	3.00	27	27.00	✓ Match
0e6d0e08329c4d0c90987d234f0983d2	6	6.00	5	5.00	5	5.00	24	24.00	✓ Match
196707e58dd34889905f2c4c3089ed5a	6	6.00	4	4.00		0.00	25	25.00	✓ Match
1d755c218f714559a57ee7df7e6b1ca0	6	6.00	6	6.00	3	3.00	26	26.00	✓ Match
21f58f38b6ac4f67acd2959072adaa7c	6	6.00	6	6.00	4	4.00	24	24.00	✓ Match
28681c16026943e68f26feaccab0907f	6	6.00	6	6.00	4	4.00	30	30.00	✓ Match
2a06fdb4f40749cb8830973a99e9672a	6	6.00	3	3.00	2	2.00	25	25.00	✓ Match
2ddc7d63c32d4606a7a45e3e70439b44	6	6.00	4	4.00	6	6.00	25	25.00	✓ Match
3a3359289baf4fea995a0e9bb2972198	6	6.00	5	5.00	4	4.00	25	25.00	✓ Match
4142f5e23db741b1af4be0287dc91c1c	6	6.00	6	6.00	1	1.00	30	30.00	✓ Match
417c8e42cba54dd0ba1ea7ee079ef87a	6	6.00	6	6.00	5	5.00	24	24.00	✓ Match
469b32f79f5f4c9ca6f05ddabc6ba6e0	6	6.00	3	3.00	3	3.00	25	25.00	✓ Match
4c4367386def4953b29bf9248de2d950	6	6.00	5	5.00	1	1.00	25	25.00	✓ Match
562b9efc809b4bd897ce0381347aaea2	6	6.00	5	5.00	6	6.00	24	24.00	✓ Match
5da599f0f0ca40a6916e28487a55e655	6	6.00	6	6.00	4	4.00	25	25.00	✓ Match
5e60c6aa3b834e44b822ea43a3efea26	6	6.00	5	5.00	5	5.00	32	32.00	✓ Match
5ed6fe9d9589427e8838b278d61732fe	6	6.00	5	5.00	1	1.00	25	25.00	✓ Match
6e92f1561dab4dc180245974f546f5b5	6	6.00	4	4.00	6	6.00	24	24.00	✓ Match
77388a70eaf14433b4efc5d01fa947bf	6	6.00	4	4.00	5	5.00	25	25.00	✓ Match
7be5dd52d7d540429299a9ee454774e7	6	6.00	4	4.00	3	3.00	27	27.00	✓ Match
Total	**674**	**674.00**	**575**	**575.00**	**421**	**421.00**	**3353**	**3,353.00**	**✓ Match**

B10. Performance and maintenance

- Raw payload and unused text columns removed or hidden; measures reuse base measures instead of repeating logic.
- Visual data volumes are small (30 days, 10 offers, 4 channels); refresh target under 3 minutes for ~306k events – confirm in refresh history.
- **Add a new offer:** add a row to Offers (offer_id, offer_type, difficulty, reward, duration, channels, OfferName) and one row per channel to OfferChannels; new events must use the same offer_id. Refresh – the offer appears in all Offer Performance visuals automatically.

B11. Requirements coverage (brief → where it is met)

Brief requirement	Where it is met
Connect to customers, events, offers, data_dictionary	Raw files cleaned into 01_Data_set_cleaned.xlsx; DataDictionary table in the model (B1, B3)
Parse JSON in events.value	offer_id, amount, reward extracted; Power Query M in B5
Standardise demographics; validate offer IDs	Unknown handling and integrity checks (B3)
Star schema with fact table and lookups	Events fact, Customers / Offers dimensions, OfferChannels bridge (B4)
Measures: sent / viewed / completed per customer and offer	B7 (17 report measures)
Segmentation and redemption by type and channel	Pages 3 and 4
Spend before and after offer completion	Page 5
Pages 1–5 as specified	Report inventory (B8)
Validate DAX for a 100-customer sample	Validation page: 0 mismatches (B9)
Optimise model; refresh under 3 minutes	B10
User guide, technical document, deployment	Parts A, B and C

Part C – Publishing to Power BI Service

Upload, permissions, scheduled refresh and sharing

C1. Publish the report

- In Power BI Desktop, open *02_Final_Customer_Offer_Engagement_Dashboard_Capstone.pbix* and press **Ctrl + S**.
- **Home → Publish** → sign in → choose the workspace (e.g. *Marketing Analytics*) → **Select**.
- When "Success!" appears, click **Open in Power BI** and check all five pages load correctly.
- The workspace now holds the **Report** and its **Semantic model**.

C2. Workspace permissions

Workspace → **Manage access** → **Add people or groups** → choose a role.

Role	Who	Can do
Admin	Project lead	Everything, including managing access
Member	Analytics team	Publish, edit and share content
Contributor	Report developers	Create and edit content; cannot share
Viewer	Marketing team / stakeholders	View and interact (slicers, filters) only

For a wider audience, publish a **Power BI App** (workspace → Create app) and give the marketing team access to the app.

C3. Scheduled refresh

Data location	Setup needed
Data file on a local PC or network drive	Install and sign in to the **On-premises data gateway** on a machine that is always on, then map the data source to it.
Data file on OneDrive for Business / SharePoint	No gateway needed: change the source in Desktop to the SharePoint/OneDrive path, republish, then sign in under Data source credentials.

- Workspace → **Semantic model** → … → **Settings** → **Gateway and cloud connections** → select the gateway or cloud connection.
- **Data source credentials** → Edit credentials → sign in.
- **Refresh** → **Configure a refresh schedule** → **Daily** (or Weekly) → time zone **(UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi** → e.g. 07:00 → tick **Send refresh failure notifications** → **Apply**.
- Test with **Refresh now** and confirm "Completed" in the refresh history.

C4. Share with stakeholders

- Open the report → **Share** → enter names or emails → **Send**; or **Copy link** and post it in the team channel.
- To share the file itself, send *02_Final_Customer_Offer_Engagement_Dashboard_Capstone.pbix*; the recipient needs Power BI Desktop (free, Windows). Include *01_Data_set_cleaned.xlsx* only if they need to refresh.
- Optional: **Subscribe** stakeholders to a weekly email snapshot of the Executive Overview page.

C5. Deployment checklist

#	Item	Done
1	All five pages checked against the validation figures (B9); no visual shows an error	☐
2	Titles, labels and axis titles reviewed on every page	☐
3	.pbix saved and published to the correct workspace	☐
4	Workspace roles assigned (Admin / Member / Viewer)	☐
5	Gateway or cloud connection configured; credentials entered	☐
6	Scheduled refresh set and a test refresh completed	☐
7	Report link or app shared with the marketing team	☐
8	This document, the presentation and the cleaned data file shared with stakeholders	☐